

Parcel Scanner Sim – AI Testing (OCR / Barcode Scanner)

1. Introduction and Test Objective

Since the main functionality of the application is scanning, testing scanners ensures that the core application flow works without any issues. The application uses the MLKit library, specifically the barcode scanner and OCR algorithms. The application was designed for employees of small shipping sorting facilities, therefore it is necessary to verify whether the scanner works under different conditions (low light, dust, different scanning angles, different input data).

2. Metrics Used in Testing

The main metric in the tests is accuracy, as we want to ensure that scanning forms and packages is precise and that the scanned data corresponds to the data on the form or label. We also want to verify whether incorrectly placed data, damaged labels, or incomplete codes/tables will prevent the application from displaying ARN or cause an exception leading to system freeze.

3. Test Scenarios

Due to two types of scanners, test cases are divided into two tables.

- Package Scanner (MLKit Barcode Scanner)

Test ID	Test Case Name	Input Data	Expected Result	Success Metric
TC-AI-01	Correct scanning of a package with a valid EAN-13 code	Small shipment with EAN-13 code belonging to InPost company (good lighting, sharp image)	Correct recognition of the EAN-13 code with proper result mapping	Accuracy > 95%
TC-AI-02	Scanning a package with an invalid EAN-13 code	Envelope with an EAN-13 code missing several digits (good lighting, sharp image)	The algorithm detects invalid EAN-13 code and shows an error or does not scan it	System does not crash, no ARN or invalid data
TC-AI-03	Scanning a package with a different barcode format	Barcode placed not on a parcel, in a completely	Algorithm does not match EAN-13 and shows a	System runs without crashes,

		different format (good lighting, sharp image)	message or ignores it	no ARN or incorrect scan
TC-AI-04	Scanning an EAN-13 code at an angle	Small shipment with InPost EAN-13 code scanned at 30–45 degree angle	Correct recognition and proper data mapping	Accuracy > 90%
TC-AI-05	Scanning in low light conditions	Small shipment with EAN-13 code in reduced lighting	Correct recognition and proper data mapping	No forced errors, no ARN or incorrect data
TC-AI-06	Scanning during movement	Small shipment with EAN-13 code, image captured while moving or shaking camera slightly	No false positives; if failure occurs, proper message or no scan	System stable, no ARN, no incorrect or false-positive scans
TC-AI-07	Scanning with flashlight enabled	Small shipment with EAN-13 code, scanned under very bright light or flashlight (partially overexposed code)	No false positives; proper handling of failure cases	System stable, no ARN or incorrect scans
TC-AI-08	Multi-code scanning (all valid codes)	Three EAN-13 codes placed in one frame, all valid	No false positives; algorithm selects first valid EAN-13 code	System scans correct code without mixing data
TC-AI-09	Multi-code scanning (EAN-13 + invalid format + QR code)	Three codes in one frame: EAN-13, invalid format, QR code	Correct EAN-13 is selected and processed	System selects correct code without interference
TC-AI-10	Multi-code scanning (all invalid codes)	Three partially obscured EAN-13 codes with missing digits	No false positives; proper error handling or no scan	System stable, no ARN, no incorrect scans

- Form Scanner (MLKit OCR)

Test ID	Test Case Name	Input Data	Expected Result	Success Metric
TC-AI-11	Correct scanning of a form with all data present	Properly displayed and structured data (good lighting, sharp image)	Data correctly recognized and mapped to result	Accuracy > 95%

TC-AI-12	Scanning a form with missing data	Form with missing fields (good lighting, sharp image)	Data recognized correctly; missing fields set to null, not filled incorrectly	Accuracy > 90%
TC-AI-13	Scanning a form in table format	Data structured in table format (e.g. payment forms, tax forms)	Data correctly extracted; layout does not affect recognition	No false positives; correct extraction
TC-AI-14	Scanning a handwritten form (legible handwriting)	Form filled with clear handwritten text (printed letters, good lighting, sharp image)	Data correctly recognized	Accuracy > 90%
TC-AI-15	Scanning a handwritten form (illegible handwriting)	Form filled with very unclear handwriting	Data partially recognized; missing fields handled properly	Data not incorrectly filled
TC-AI-16	Scanning data from screen (screenshot/PDF)	Data displayed on screen (computer/phone)	Data correctly extracted	Accuracy > 90%, no false positives
TC-AI-17	Scanning a tilted form	Form photographed at 30–45 degree angle	Correct recognition with deskew applied	Accuracy > 90%, no false positives
TC-AI-18	Scanning in poor conditions (wet paper)	Form with visible water damage but readable data	Data correctly recognized	Accuracy > 90%
TC-AI-19	Scanning overlapping documents	Multiple documents partially overlapping in frame	System correctly distinguishes target form	Accuracy > 90%